WORKED SOLUTION

STEP III 2022, Question 1

Pure • Roots of Polynomials

STEP III 2022, Question 1

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Substituting the parametrisation into the circle's equation turns "where do the curves meet?" into "what are the roots of a quartic?", and from then on everything is done with the relationships between roots and coefficients, without ever solving the quartic. The useful extra trick is that the reciprocals $1/t_i$ are the roots of the same quartic with its coefficients written in reverse order. For the final part, "touching" means a repeated root.

Solution

Part (i)

The point with parameter t lies on C_2 exactly when

$$(ct-a)^2+\left(\frac{c}{t}-b\right)^2=r^2.$$

Expanding gives $c^2t^2-2act+a^2+\dfrac{c^2}{t^2}-\dfrac{2bc}{t}+b^2=r^2$. Since $t\neq 0$ we may multiply by t^2 without gaining or losing solutions:

$$c^2t^4-2act^3+(a^2+b^2-r^2)t^2-2bct+c^2=0.$$

So each t_i is a root of $(*)$. Different values of t give different points (they have different x-coordinates ct), so the four points P_i give four distinct parameters. A quartic has at most four roots, so $t_1,\dots,t_4$ are exactly the roots of $(*)$.

Part (ii)

Dividing $(*)$ by c^2, the sums of the roots taken one and two at a time are

$$\sum t_i=\frac{2a}{c},\qquad \sum_{i<j}t_it_j=\frac{a^2+b^2-r^2}{c^2}.$$

Using $(\sum t_i)^2 = \sum t_i^2 + 2\sum_{i<j} t_i t_j$,

$$\sum_{i=1}^{4} t_i^2 = \frac{4a^2}{c^2} - \frac{2(a^2+b^2-r^2)}{c^2} = \frac{2}{c^2}(a^2 - b^2 + r^2),$$

as required.

? **Ask yourself** How can we get at $\sum 1/t_i^2$ without a messy common denominator?

Put $u = 1/t$ in $(*)$. No root is zero (the constant term is $c^2 \neq 0$), so we can divide $(*)$ by t^4:

$$c^2 - 2acu + (a^2+b^2-r^2)u^2 - 2bcu^3 + c^2u^4 = 0,$$

that is, $c^2u^4 - 2bcu^3 + (a^2+b^2-r^2)u^2 - 2acu + c^2 = 0$. Its roots are the $1/t_i$. This is $(*)$ with a and b swapped, so the same calculation gives

$$\sum_{i=1}^{4} \frac{1}{t_i^2} = \frac{2}{c^2}(b^2 - a^2 + r^2).$$

Part (iii)

The point P_i is $(ct_i,\, c/t_i)$, so $OP_i^2 = c^2t_i^2 + \dfrac{c^2}{t_i^2}$. Summing and using part (ii),

$$\begin{aligned}\sum_{i=1}^{4} OP_i^2 &= c^2\sum t_i^2 + c^2\sum \frac{1}{t_i^2} \\ &= 2(a^2 - b^2 + r^2) + 2(b^2 - a^2 + r^2) \\ &= 4r^2,\end{aligned}$$

which is the required result. Notice that the centre (a, b) has completely disappeared.

Part (iv)

? **Ask yourself** What does "touch" mean in terms of the roots of $(*)$?

Let $f(t) = (ct - a)^2 + (c/t - b)^2 - r^2$, the squared distance from the moving point to the centre, minus r^2. The curve touches the circle at parameter t_0 when $f(t_0) = 0$ and $f'(t_0) = 0$: the point is on the circle and, to first order, its distance from the centre is not changing, so the curve's direction there is perpendicular to the radius, which is the circle's tangent direction. (This uses the fact that the velocity $(c, -c/t^2)$ is never zero.) The quartic in $(*)$ is $t^2 f(t)$, and since $t_0 \neq 0$, f has a double root at t_0 if and only if $(*)$ does.

So if C_1 and C_2 touch at two distinct points, with parameters $\alpha \neq \beta$, then α and β are both double roots of $(*)$. That accounts for all four roots, so

$$c^2t^4 - 2act^3 + (a^2+b^2-r^2)t^2 - 2bct + c^2 = c^2(t-\alpha)^2(t-\beta)^2.$$

Product of the roots. Comparing constant terms, $c^2\alpha^2\beta^2 = c^2$, so $(\alpha\beta)^2 = 1$ and $\alpha\beta = \pm 1$.

Sum of the roots. Comparing t^3 terms, $2(\alpha+\beta) = \dfrac{2a}{c}$, so $\alpha+\beta = \dfrac{a}{c}$.

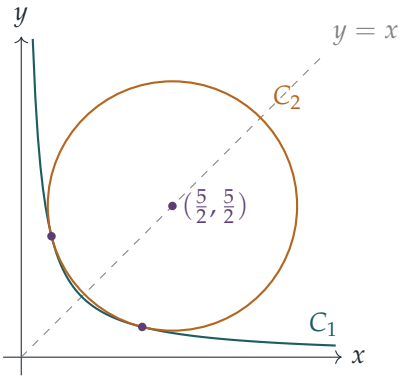
Sum of products of three. Comparing t terms, $2\alpha\beta(\alpha+\beta) = \dfrac{2b}{c}$, so $\alpha\beta(\alpha+\beta) = \dfrac{b}{c}$.

Dividing the last two (or substituting the second into the third) gives $b = \alpha\beta\, a$. Hence

$$\alpha\beta = 1 \implies b = a, \qquad \alpha\beta = -1 \implies b = -a,$$

so the centre (a, b) lies on $y = x$ or on $y = -x$. (If $a = 0$ then $b = 0$ too, and the centre is the origin, which lies on both lines.)

The figure shows the case $c = 1$, $\alpha = 2$, $\beta = \frac{1}{2}$. Matching coefficients gives $a = b = \frac{5}{2}$ and $r^2 = \frac{17}{4}$, and the circle touches the hyperbola at $(2, \frac{1}{2})$ and $(\frac{1}{2}, 2)$.



Checks

- In part (iii), take the circle $x^2 + y^2 = r^2$ centred at the origin with $c = 1$ and $r^2 > 2$. By symmetry the four points are $(p, q), (q, p), (-p, -q), (-q, -p)$, each at distance r, so the total really is $4r^2$.
- In the example in the figure, the squared distance from $(\frac{5}{2}, \frac{5}{2})$ to $(2, \frac{1}{2})$ is $\frac{1}{4} + 4 = \frac{17}{4} = 4.25$, as it should be. The radius direction $(-\frac{1}{2}, -2)$ is perpendicular to the curve's direction $(1, -\frac{1}{4})$ there, confirming tangency.
- The formulas in part (ii) were confirmed by computer for a few thousand random circles, by solving the quartic numerically.

Common slips

- Getting the sign of $\sum t_i t_j$ wrong. For $t^4 + pt^3 + qt^2 + \ldots$ the sum of the roots is $-p$ but the sum of the pairwise products is $+q$.

- In part (iv), using only the product of the roots. That gives $\alpha\beta = \pm 1$, which on its own says nothing about (a, b). One more coefficient comparison is needed to bring in a and b.
- Not saying why "touching" gives a double root. The reason is the condition $f(t_0) = f'(t_0) = 0$, together with $t_0 \neq 0$.

Beyond STEP

The relations between the roots and the coefficients are the elementary symmetric functions. A classical theorem says that *every* symmetric polynomial in the roots, such as $\sum t_i^2$ or $\sum t_i^{-2}$, can be written in terms of the coefficients. The reversal trick $u = 1/t$ is a first glimpse of projective ideas, and the fact that a conic and a circle meet in at most four points is a special case of Bézout's theorem on intersecting plane curves. For more on polynomials and their roots, see Allenby, *Rings, Fields and Groups*; for practice in this style, Siklos, *Advanced Problems in Mathematics*.

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